Structure-Guided Feature Transform Hybrid Residual Network for Remote Sensing Object Detection

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***Abstract*— Object detection in remote sensing imagery (RSI) is a fundamental task for Earth monitoring. Objects captured from the bird's eye view perspective in RSI can appear as multiscale in arbitrary orientations, most of which are small and dense. In specific, vehicles or ships only occupy a dozen pixels in the image, but are surrounded by roads and seas, which occupy thousands of pixels and comprise overwhelmingly dominant of all pixels. Although a large number of common object detection methods have been proposed, most of them cannot detect small and dense objects accurately because none of them has paid enough attention to the unique characteristic of RSI. In this work, we propose a novel structure-guided feature transform hybrid residual (SGFTHR) network, which can conquer the low performance of detection of objects at different scales, especially for small and dense objects, in an anchor-free manner. The structure-guided feature transform (SGFT) module is promoted to extract discriminative structural information and guide this information into high-level contextual feature maps, preventing the important low-level spatial and structural information from being lost when the network goes deeper. Furthermore, the hybrid residual (HR) module is embedded in the backbone to acquire multiscale features in a novel hybrid hierarchical residual-like manner. Extensive experiments are performed on the HRRSD and NWPU VHR-10 datasets to evaluate the performance of the SGFTHR network, which demonstrates that our SGFTHR network achieves state-of-the-art detection accuracy with high efficiency and robustness. Specifically, 4.12% improvements in mean average precision (mAP) on the HRRSD dataset compared with baseline powerfully demonstrate the effectiveness and superiority of the SGFTHR network.**

***Index Terms*— Discriminative structural information, hybrid hierarchical residual-like manner, multiscale features, object detection, remote sensing imagery (RSI).**

I. Introduction

OBJECT detection of remote sensing imagery (RSI), which calculates the specific location and assign a unique category label for an object, is widely applied, such as vehicle detection [1]–[4], building detection [5]–[8], and ship detection [9]–[11]. However, due to the bird's eye view perspective, objects in RSI are quite different from natural scene imagery in terms of several specific characteristics, including the randomness of distribution, the complexity of background, the variability of light and weather, and the disunity in ground sample distances, to name a few. Hence, the common object detection methods perform poorly when directly applied to remote sensing object detection.

Inspired by the awesome feature representation capability, numerous researches based on deep convolutional neural networks (DCNNs) have been generated to improve the accuracy and efficiency of detection. Current object detectors consist of two mainstreams: two-stage and one-stage. Two-stage detectors usually can generate high-quality proposals through a region proposal network in the first stage, which are further utilized to classify and locate objects in the second stage. Two-stage detectors acquire great achievement and performance in precision but are resource-consuming and time-consuming, for instance, R-CNN [12], Fast R-CNN [13], Faster R-CNN [14], Mask R-CNN [15], and R-FCN [16]. Meanwhile, one-stage detectors calculate specific locations and categories of objects directly through an efficient network without region proposal process, which achieve great efficiency, such as YOLO series [17]–[20], SSD [21], and DSSD [22]. Particularly, part of one-stage detectors is proposal-free and anchor-free, which not only simplifies the training process but also is closer to real-time detection, such as CornerNet [23], CenterNet [24], and fully convolutional one-stage object detector (FCOS) [25]. However, most detectors do not pay enough attention to the unique characteristic of RSI, such as the dense arrangement of small objects and the complexity of the background, which

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brings great difficulties to remote sensing object detection. Thus, the detectors cannot be applied directly to address the object detection tasks in RSI. On the one hand, the objects captured from the top view can appear as multiscale in arbitrary orientations, such as bridges between two sides and airplanes in an airport. Additionally, the variability of lighting and weather, the arbitrariness of aspect ratio, and the complexity of the background bring a challenge for improving the object detection performance of RSI. On the other hand, vehicles and ships captured from overhead are small and dense, which only occupy a dozen pixels in RSI. However, these small objects are surrounded by roads and seas, which occupy thousands of pixels in the image and comprise overwhelmingly dominant of all pixels.

The straightforward manner is to exploit these kinds of detectors directly, however, according to the aforementioned characteristic, the existing remote sensing detection models can be roughly classified into the following aspects [26]: template matching-based [27], knowledge-based [28], object-based [29], machine learning-based methods [30], and other variants. In the template matching-based methods, the best matches will be founded by matching the image and template generated through hand-crafting or learned from the training set at each possible position according to the minimum distortion. The knowledge-based methods generally transform object detection tasks into hypotheses testing tasks via establishing various knowledge. The object-based methods segments image into homogeneous regions with scale, shape, and compactness as the selection standard and detect objects in a classification manner. Motivated by the powerful capability of feature representation, deep learning has attracted extensively researchers' attention, and attained significant improvements in detection performance, such as [31]–[36]. Nevertheless, none of them has put enough emphasis on the prior knowledge of the input images. The discriminative structural information, containing abundant, detailed, and characteristic shape features, is of critical importance to boost the performance in detecting the small and dense objects but is prone to weaken even disappear when networks go deeper [37].

In order to conquer the aforementioned issues, a novel one-stage network called structure-guided feature transform hybrid residual (SGFTHR) network is proposed. Through comprehensively utilizing the discriminative structural information and the multiscale features, our SGFTHR network is capable of precisely detecting objects with the dense arrangement in RSI. The main contributions of this article are specified as follows.

1) In detail, we propose a novel SGFTHR network for detecting objects, which can overcome the low performance of detection of targets with complex distribution at different scales in an anchor-free manner. Extensive experiments on both HRRSD and NWPU VHR-10 datasets demonstrate that our proposed SGFTHR network can achieve state-of-the-art performance compared with previous algorithms designed for natural images and RSI, respectively.
2) The structure-guided feature transform (SGFT) module is proposed to prevent the important low-level spatial and structural information from being lost when the network goes deeper. Specifically, the discriminative structural features are captured from the input image through the structure tensor extractor and guided into high-level contextual feature maps to increase informative representations, which is beneficial for boosting accuracy.
3) An innovative hybrid residual (HR) module is also invented to acquire multiscale features in a novel hybrid hierarchical residual-like manner. To be specific, the HR module consists of four different kernel sizes within a residual block, which enhances the model's ability to represent multiscale features at a granular level.

The remainining of our article is constructed as follows. Section II reviews the related works of object detection. Section III introduces our proposed object detection framework in detail. The experimental results of the proposed method are given in Section IV. Finally, the conclusion of this article is drawn in Section V.

II. Related Work

Object detection with various applications has been received increasing interest. In this section, the SOTA object detection frameworks in both natural imagery and RSI are reviewed in detail. Besides, the superiority of the proposed framework compared with other relevant methods is also clarified and emphasized.

A. Object Detection in Natural Imagery

In the past decades, a large amount of deep learning-based methods designed for object detection has successfully achieved great significance in natural imagery. Specifically, the existing machine learning-based object detection methods can be roughly categorized into two streams based on whether generating region proposals or not, which are two-stage methods and one-stage methods. In the two-stage fashion, the first stage aims to generate a series of high-quality candidate region proposals for possible objects. The second stage focuses on classifying the candidate region proposals acquired from the first stage into object categories and further refining the coordinates of the bounding boxes. Girshick *et al.* [12] propose a representative region proposal-based method named R-CNN that achieves breakthrough performance progress compared with all previous methods. In order to address the low efficiency of R-CNN and acquire better accuracy, Ren *et al.* [14] propose Faster R-CNN that only needs to calculate once to generate high-quality region proposals through a CNN model. Additionally, Mask R-CNN [15] adds a parallel branch to predict an object mask for detecting bounding box on the basis of Faster R-CNN for the purpose of further boosting the performance of Faster R-CNN. Meanwhile, the detection task is simplified as a regression problem in the one-stage methods, which is much more concise and efficient than the two-stage methods. Bounding boxes and class probabilities are predicted through a single CNN backbone in one evaluation in the one-stage method. The representative one-stage methods are YOLO series [17]–[20], SSD [21], RetinaNet [38]. With the purpose of boost both the speed and accuracy, a novel

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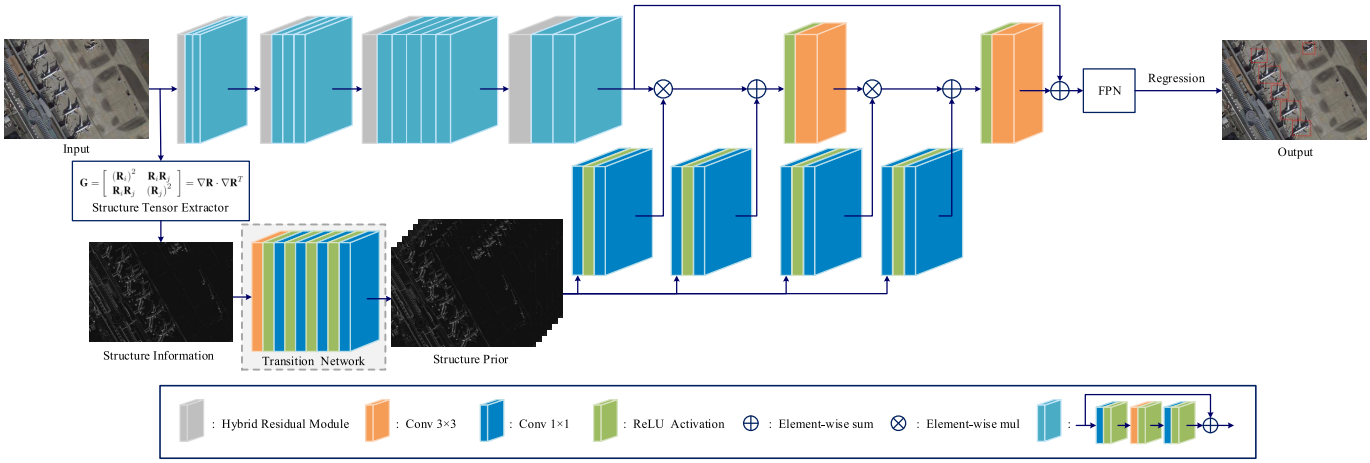

Fig. 1. Pipeline of our SGFTHR network.

anchor free object detection paradigm is proposed, including CornerNet [23], CenterNet [24], FCOS [25], which eliminates the predefined set of anchor boxes and avoid a large number of parameters and complicated calculation. ConerNet detects objects as paired corners, while FCOS detects objects in the scale of a pixel, similar to semantic segmentation.

B. Object Detection in RSI

In recent years, extensive research has been devoted to object detection in RSI. Driven by the excellent performance of the classic two-stage method (Faster R-CNN) in natural object detection, numerous research adopts Faster R-CNN as the baseline to boost the detection performance in RSI. For instance, Li *et al.* [39] introduced additional multiangle anchors into the RPN network based on Faster R-CNN in view of the multiscale and multiangle characteristics of remote sensing objects, which can deal with the problem of rotation variations. Additionally, aiming at handling the problem of rotation variations, a double-channel feature fusion network is proposed to extract local and contextual information. In order to handle the challenge between translation-invariance in the classification stage and translation-variance in the object detection stage, Zhong *et al.* [40] present a position-sensitive balancing framework that makes full use of the fully convolutional network (FCN). Guo *et al.* [41] propose a unified multiscale convolutional neural network with the purpose of precisely detecting objects with multiscale, especially small objects in the high-resolution RSI. Recently, aiming at achieving real-time object detection, some research develops one-stage methods from natural images to RSI. For example, Liu *et al.* [42] embed the rotatable bounding box into the SSD framework replacing the traditional oriented bounding box. Liu *et al.* [43] adopt YOLOv2 [18] as baseline to directly predict arbitrary oriented ships.

Compared with these existing detection methods, the FCOS, selected as the baseline of our proposed SGFTHR network, is an anchor-free and proposal-free detector and detects objects in a per-pixel prediction manner, which provides a vital foundation for the detection of dense objects in RSI.

III. Methodology

The characteristics of the objects in RSI are the large difference in distribution and the dense arrangement of small-scale objects, which hinder the detection accuracy from further improving. In order to boost the detection performance of small objects, there is a large amount of research that only concentrates on context information. Since small objects occupy only dozens of pixels, the detection performance should be greatly enhanced by adopting discriminative structural information as the auxiliary information. The discriminative structural information, existing in the front sections of the whole network and containing both sufficient detailed information and more semantics of small objects, is of great significance for small object detection. In this article, we propose an anchor-free framework, namely, an innovative SGFTHR network, to make full use of the discriminative structural information and multiscale features. The overall architecture of our network is shown in Fig. 1, which consists of the SFTG module and the HR module. The thickness of the convolutional layer in the figure represents the number of channels. The thicker the convolutional layer, the bigger the number of channels.

A. Framework of SGFTHR Network

In the DCNNs, deep feature maps contain rich semantic information but poor detailed spatial features, because the spatial information in shallow feature maps is prone to be lost when shallow feature maps are not continuously propagated or concatenated via skip connections. It has been demonstrated in [44] that the object size has a significant effect on visual perception and more attention will be paid to small objects. Since the object with a small geometry size only occupies a small part of the entire scene, elaborate structural information, such as corners and edges, is essential for characterizing small objects. In order to boost the detection accuracy of small objects in RSI, the distinctive spatial information in objects with multiscale, especially for small-scale, should be utilized as adequately as possible. A general way is to utilize augmentation for small-scale objects. In detail, any objects are

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copied from their original location to other positions, which can advance the small objects' contribution to computing the training loss. However, this kind of method is time-consuming, and the copy of objects may cover the other objects.

Through the analysis above, we propose the SGFTHR network for object detection in RSI. The SGFTHR network constructs the abundant distinctive structural features from shallow feature maps and guides them into the posterior convolutional layers of the network through the proposed SGFT module. Moreover, objects in RSI vary in distribution and size. Considering the difficulty of detecting small objects and the necessity of multiscale features in remote sensing object detection, we exploit an innovative HR module to utilize the multiscale features and enhance the spatial details of the small objects. Next, the SGFT module and the HR module will be introduced in detail, respectively.

B. SGFT Module

As we all know, FCOS not only achieves high detection efficiency because of its anchor-free and proposal-free but also solves detection tasks in a per-pixel approach, which is similar to dense prediction missions such as semantic segmentation. However, FCOS does not perform well on objects with small geometry size in RSI, since the layers among the detectors of FCOS cannot extract or preserve enough discriminative structural information of small objects. Hence, a novel SGFT module is proposed to embed the characteristic structural features of tiny objects from shallow layers into hierarchial deep layers.

To acquire as much discriminative structural information as possible, the structure tensor extractor is exploited. The working principle and algorithm details are as follows.

Given a remote sensing image $\mathbf{R}(i, j)$ as the input, we first get the changes c produced by translation $(\Delta i, \Delta j)$, which can be formulated as

$$c = \sum_{(i,j)} s(i, j)[\mathbf{R}(i + \Delta i, j + \Delta j) - \mathbf{R}(i, j)]^2 \tag{1}$$

where the value of $(\Delta i, \Delta j)$ has the following four possibilities, $\{(0, 1), (1, 0), (1, 1), (-1, 1)\}$ and $s(i, j)$ denotes the shift window. With the first-order Taylor series

$$\mathbf{R}(i + \Delta i, j + \Delta j) = \mathbf{R}(i, j) + \mathbf{R}\Delta i + \mathbf{R}\Delta j + O\left(\Delta i^2 + \Delta j^2\right) \tag{2}$$

the change c is computed by

$$c = \sum_{(i,j)} s(i, j)\left[\mathbf{R}_i \Delta i + \mathbf{R}_j \Delta j + O\left(\Delta i^2 + \Delta j^2\right)\right]^2 \tag{3}$$

where $\mathbf{R}_i = \partial\mathbf{R}/\partial i$ and $\mathbf{R}_j = \partial\mathbf{R}/\partial j$ represent the horizontal and vertical gradient vectors, respectively. When the translation is small, the calculation of the change c can be refined into

$$c = [\Delta i, \Delta j]\ \mathbf{S_t}\ [\Delta i, \Delta j]^T \tag{4}$$

where the matrix $\mathbf{S_t}$ denotes the structure tensor, which is calculated as follows:

$$\mathbf{S_t} = \begin{bmatrix} \sum_{(i,j)} s(i, j)(\mathbf{R}_i)^2 & \sum_{(i,j)} s(i, j)\mathbf{R}_i\mathbf{R}_j \\ \sum_{(i,j)} s(i, j)\mathbf{R}_i\mathbf{R}_j & \sum_{(i,j)} s(i, j)\left(\mathbf{R}_j\right)^2 \end{bmatrix}. \tag{5}$$

From the characteristics of the semidefinite matrix $\mathbf{S_t}$, the structure tensor $\mathbf{S_t}$ can be described as

$$\mathbf{S_t} = \begin{bmatrix} \lambda_1 & \lambda_2 \end{bmatrix} \begin{bmatrix} \rho_1 & 0 \\ 0 & \rho_2 \end{bmatrix} \begin{bmatrix} \lambda_1 & \lambda_2 \end{bmatrix}^T \tag{6}$$

where ρ_1 and ρ_2 refers to the nonnegative eigenvalues, which contain the discriminative spatial structure information. λ_1 and λ_2 refers to the corresponding eigenvectors. $\rho_1 \approx \rho_2 \approx 0$ represent the flat area. When $\rho_1 > \rho_2 \approx 0$, this represents the edge area. $\rho_1 \geq \rho_2 > 0$ mean the corner area [45]. It can be easily analyzed from the earlier discussion that the relatively large eigenvalues represent high-frequency details. Based on the conclusion discussed above, we can capture the structure information and utilize it to detect small objects as accurately as possible by choosing larger eigenvalues. Structure tensor $\mathbf{G}$ is defined as follows:

$$\mathbf{G} = \begin{bmatrix} (\mathbf{R}_i)^2 & \mathbf{R}_i\mathbf{R}_j \\ \mathbf{R}_i\mathbf{R}_j & \left(\mathbf{R}_j\right)^2 \end{bmatrix} = \nabla\mathbf{R} \cdot \nabla\mathbf{R}^T \tag{7}$$

where $\cdot$ denotes matrix multiplication and $\nabla\mathbf{R} = [\mathbf{R}_i, \mathbf{R}_j]^T$ denotes gradient operator. Afterward, the transition network, composed of five convolutional layers, transforms the structure tensor $\mathbf{G}$ into the intermediate discriminative structural feature maps $\mathbf{F}$. In order to prevent the interference of different regions in one image from producing, the receptive field of the transition network is limited through setting 1×1 kernels for convolutional layers. The intermediate structural features $\mathbf{F}$ can be described as

$$\mathbf{F} = f_{\text{cn}}(\mathbf{G}) \tag{8}$$

where f_{cn} refers to the transition network. The proposed SGFT learns an adaptive spatial mapping function f_{asm} that takes discriminative structural feature maps $\mathbf{F}$ as input and generates a pair of adaptive guidance parameters (θ_i, β_i), where i, $j \in \{1, 2\}$. The process can be formulated as follows:

$$(\theta_i, \beta_i) = f_{\text{asm}}(\mathbf{F}). \tag{9}$$

Given the input features maps $\mathbf{F}_m$, the proposed SGFT conditioned on the discriminative structural information as the prior knowledge of input images applies an affine transformation to $\mathbf{F}_m$ by the pair of adaptive guidance parameters (θ_i, β_i). The proposed SGFT transfers the intermediate structural features into high-level contextual layers through the affine transformation, which is defined as follows:

$$\text{SGFT}(\mathbf{F}_m|\theta_i, \beta_i) = \theta_i \odot \mathbf{F}_m + \beta_i \tag{10}$$

where F_m refers to the feature maps that has the same dimension with θ_i and β_i, and $\odot$ denotes the element-wise multiplication.

After the previous process, the discriminative structural information contained in shallow layers has been guided

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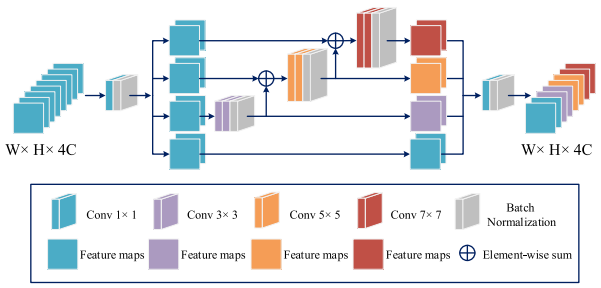

Fig. 2. Overall structure of the HR module.

into the high-level convolutional layers near the detector, which will extremely enhance the accuracy and efficiency of detection. Furthermore, we utilize skip connection to fuse the features from different layers and to refine the output, so that the model can converge faster.

C. HR Module

Existing object detectors developed in RSI are prone to be influenced by complex backgrounds and diverse distribution, especially dense arrangements of small objects [46]. Although FCOS jointly addresses the detection problem in the segmentation way, it is still a challenging task to advance the detection performance of objects with different scales, especially small targets which are ubiquitous in RSI. With the consideration of the significance of multiscale features for detecting objects with different scales, especially small objects with the dense arrangement, we propose a novel HR module embedded in the backbone network of our SGFTHR network by replacing the residual block to achieve better multiscale feature representation. By constructing hybrid hierarchical residual-like connections with different kernel sizes in a single residual block, the HR module represents features in multiple scales at a granular level.

The overall structure of our proposed HR is shown in Fig. 2. The W, H, and $4C$ denote the width, height, and channels of the feature maps, respectively. This module consists of four branches, which not only make the feature map contain multiscale information but also enhance the spatial details in the feature map, especially for small and dense objects. After the 1×1 convolution and batch normalization, the input feature map is evenly divided into four feature map subsets, denoted by $\mathbf{In}_i$, where $i \in \{1, 2, 3, 4\}$. Each of the feature map subsets has the same spatial size as the original input feature map but a quarter of number of channels. $\mathbf{In}_2$, $\mathbf{In}_3$, $\mathbf{In}_4$ have two consecutive 3×3, 5×5, 7×7 convolution layers and batch normalization referred to $\mathbf{C}_i(\cdot)$, respectively, except for $\mathbf{In}_1$. The output of $\mathbf{C}_i(\cdot)$ is defined as $\mathbf{Out}_i$. We add the feature map subsets $\mathbf{In}_i$ to the output of $\mathbf{C}_{i-1}(\cdot)$ and then feed it into $\mathbf{C}_i(\cdot)$. More precisely, the $\mathbf{Out}_i(\cdot)$ can be formulated as follows:

$$\mathbf{Out}_i = \begin{cases} \mathbf{In}_i, & i = 1 \\ \mathbf{C}_i(\mathbf{In}_i), & i = 2 \\ \mathbf{C}_i(\mathbf{In}_i + \mathbf{Out}_{i-1}), & i = 3, 4. \end{cases} \tag{11}$$

Each time the feature map subset goes through the two consecutive convolution layers with different kernel sizes and batch normalization, the output feature map can contain a larger receptive field than the input feature map. On account of the combinatorial explosion effect, the output feature map of the HR module comprises a different number and various combinations of receptive field scales.

In our proposed HR module, each subset is constructed in a hybrid hierarchical residual-like manner with different kernel sizes. Hence, the output of the HR module contains features at different scales. By fusing these features with different scales, the HR module promotes the number of scales that the output features can represent, which can not only improve the multiple available receptive fields at a more granular level but also advance the performance of detecting objects with multiscale while being more efficient and effective. Being dealt with multiscale ways, the subsets not only capture both global and local information but also pay attention to the spatial details of the object with small geometry size. Additionally, the split and concatenation operation can improve the efficiency of processing features.

TABLE I
ABLATION EXPERIMENTS ON HRRSD DATASET

	None	SGFT	HR	SGFTHR
Airplane	0.9607	0.9648	0.9710	**0.9732**
Baseball Diamond	0.8682	0.8762	0.8897	**0.8956**
Basketball Court	0.4857	0.5508	0.5476	**0.5777**
Bridge	0.8605	0.8876	0.9044	**0.9047**
Crossroad	0.8899	0.8958	0.9028	**0.9269**
Ground Track Field	0.9501	0.9553	0.9580	**0.9684**
Harbor	0.9487	0.9606	0.9603	**0.9615**
Parking Lot	0.4995	0.5444	0.5609	**0.6038**
Ship	0.8763	0.9017	0.9093	**0.9121**
Storage Tank	0.9444	0.9457	0.9432	**0.9501**
T Junction	0.6384	0.6701	0.6910	**0.7421**
Tennis Court	0.8177	0.8332	0.8288	**0.8493**
Vehicle	0.9537	0.9601	0.9641	**0.9644**
Mean AP	0.8226	0.8420	0.8486	**0.8638**

IV. EXPERIMENTS

In order to evaluate the validity of our proposed SGFTHR network, a definite description of the experimental details is provided, including datasets, experimental setup, evaluation metrics, experimental results, and analysis with comprehensive discussion. Several prevalent and progressive remote sensing detection methods are utilized as comparison, such as Faster R-CNN [14], Mask R-CNN [15], RetinaNet [38], FCOS [25], and CenterNet [24]. The remote sensing object detection methods are not included in the comparison methods, because the code is not open source, we cannot compare the SGFTHR network fairly with them.

A. Datasets

Aiming at advancing the development of object detection in RSI, numerous public datasets [47]–[51], are born one after another for further research. To verify and evaluate the effectiveness and superiority of our proposed SGFTHR network, the following two datasets, widely utilized with diversity, are employed to carrying out experiments.

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TABLE II
COMPARISON OF THE PERFORMANCE OF OTHER METHODS ON HRRSD DATASET

	Faster R-CNN	Mask R-CNN	RetinaNet	FCOS	CenterNet	SGFTHR
Airplane	0.9669	**0.9771**	0.9737	0.9607	0.9487	0.9732
Baseball Diamond	0.8777	0.8690	0.8733	0.8682	0.8610	**0.8956**
Basketball Court	0.5107	0.5555	0.5498	0.4857	0.5172	**0.5777**
Bridge	0.8316	0.8255	0.8513	0.8605	0.8522	**0.9047**
Crossroad	0.8954	0.8958	0.8858	0.8899	0.9258	**0.9269**
Ground Track Field	0.9668	0.9585	**0.9710**	0.9501	0.9515	0.9684
Harbor	0.9326	0.9395	0.9168	0.9487	0.8960	**0.9615**
Parking Lot	0.5535	0.5913	0.4864	0.4995	0.5595	**0.6038**
Ship	0.8691	0.8794	0.8688	0.8763	0.8242	**0.9121**
Storage Tank	0.9306	0.9337	0.9396	0.9444	0.9270	**0.9501**
T Junction	0.7087	0.7217	0.7149	0.6384	**0.7661**	0.7421
Tennis Court	0.8656	**0.8851**	0.8818	0.8177	0.8463	0.8493
Vehicle	0.8937	0.9106	0.9140	0.9537	0.8528	**0.9644**
Mean AP	0.8310	0.8417	0.8329	0.8226	0.8253	**0.8638**

TABLE III
COMPARISON OF THE PERFORMANCE OF OTHER METHODS ON NWPU VHR-10 DATASET

	Faster R-CNN	Mask R-CNN	RetinaNet	FCOS	CenterNet	SGFTHR
Airplane	0.7801	**0.8083**	0.7810	0.7784	0.7594	0.7821
Ship	0.6630	0.6702	0.6560	0.6689	0.6570	**0.6718**
Storage Tank	0.8488	0.8422	0.8485	0.8503	0.8312	**0.8649**
Baseball Diamond	0.9515	0.9706	0.9684	0.9416	0.9254	**0.9749**
Tennis Court	0.7657	**0.8771**	0.6689	0.8175	0.7275	0.8581
Basketball Court	0.4338	0.4573	0.4485	0.4019	0.4386	**0.4615**
Ground Track Field	0.8668	0.8574	**0.8877**	0.8719	0.8698	0.8745
Harbor	0.3664	0.3680	0.3560	0.3754	0.3540	**0.3791**
Bridge	0.4900	0.4764	**0.5267**	0.4958	0.4859	0.5099
Vehicle	0.7973	0.7691	0.6889	0.8735	0.7289	**0.9333**
Mean AP	0.6964	0.7097	0.6831	0.7075	0.6778	**0.7310**

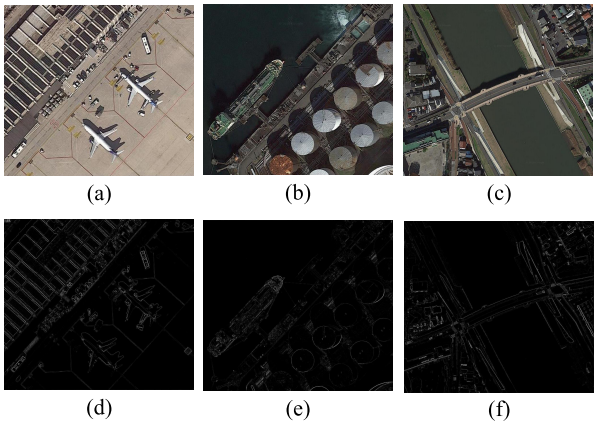

Fig. 3. (a)–(c) Origin images from the HRRSD dataset. (d)–(f) Visualization results obtained by the structure tensor extractor.

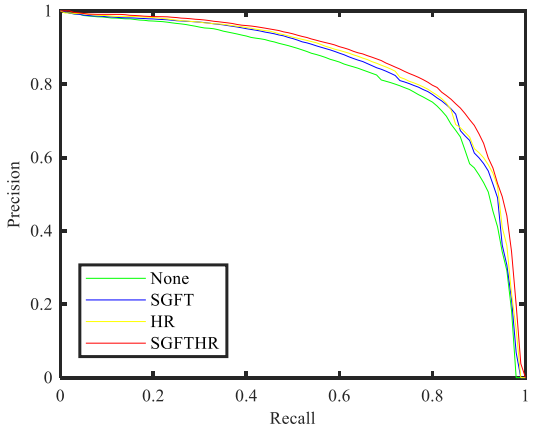

Fig. 4. PRCs of different methods on the HRRSD dataset.

1) HRRSD Dataset: It contains 21 761 images from Google Earth satellite imagery, whose spatial resolution varies from 0.15 to 1.2 m. The fully annotated HRRSD dataset [52] is a 13-class high-resolution remote sensing object detection dataset with diversity. Thirteen categories of objects and the proportion of each category are ship (8.90%), bridge (8.20%), ground track field (6.70%), storage tank (7.90%), basketball court (6.95%), tennis court (7.71%), airplane (8.79%), baseball diamond (7.24%), harbor (6.70%), vehicle (8.53%), crossroad (7.22%), T junction (7.93%), and parking lot (8.90%). Each category has 3700–5000 instances evenly distributed.

2) NWPU VHR-10 Dataset: With the purpose of verifying the robustness of our proposed SGFTHR network, a series of experiments is carried out on the NWPU VHR-10 dataset [53], which contains a total of 800 images with very high resolution. The dataset is related to ten categories. The ten

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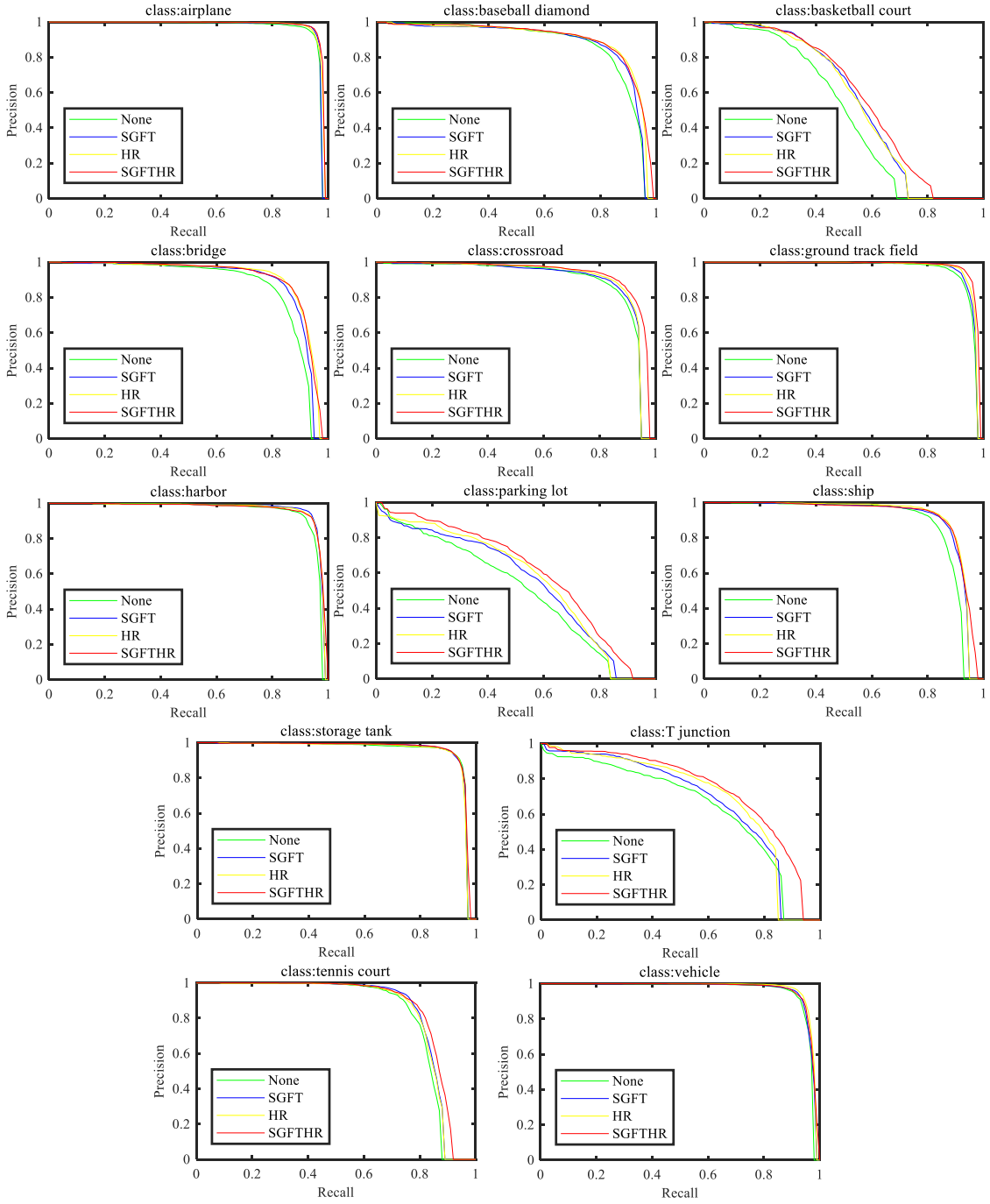

Fig. 5. PRCs of different categories on the HRRSD dataset.

categories of objects and the proportion of each category are airplane (19.43%), ship (7.75%), storage tank (16.81%), baseball diamond (10.01%), tennis court (13.45%), basketball court (4.08%), ground track field (4.18%), harbor (5.57%), bridge (3.18%), and vehicle (15.35%). In this dataset, 715 images are acquired from Google Earth with the spatial resolution varying from 0.5 to 2 m, and 85 images are acquired from Vaihingen data with a spatial resolution of 0.08 m.

B. Experimental Setup

With the consideration of the difficulty of detection and the characteristics of objects in RSI, FCOS [25] is selected as the baseline of our proposed model because the way of detecting objects in FCOS is a per-pixel prediction process, detecting dense objects without anchor boxes. Besides, ATSS [54] is employed as an automatic and adaptive way to select positive and negative training samples based on object characteristics in RSI. The ResNet-50 [55] is selected as the backbone network for improving the accuracy while maintaining a satisfactory efficiency in our proposed SGFTHR network. Through resizing, the size of input images vary from 800 $\times$ 800 to 1333 $\times$ 1333. Specifically, we adopt stochastic gradient descent (SGD) in the training processing to optimize our model. Our network is trained on a batch size of six images.

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Fig. 6. Example results on the HRRSD test split (86.38% mAP). The first two rows correspond to success cases. The third rows indicate a few missed objects and FPs. (categories: 1-airplane, 2-baseball diamond, 3-basketball court, 4-bridge, 5-crossroad, 6-ground track field, 7-harbor, 8-parking lot, 9-ship, 10-storage tank, 11-T junction, 12-tennis court, and 13-vehicle).

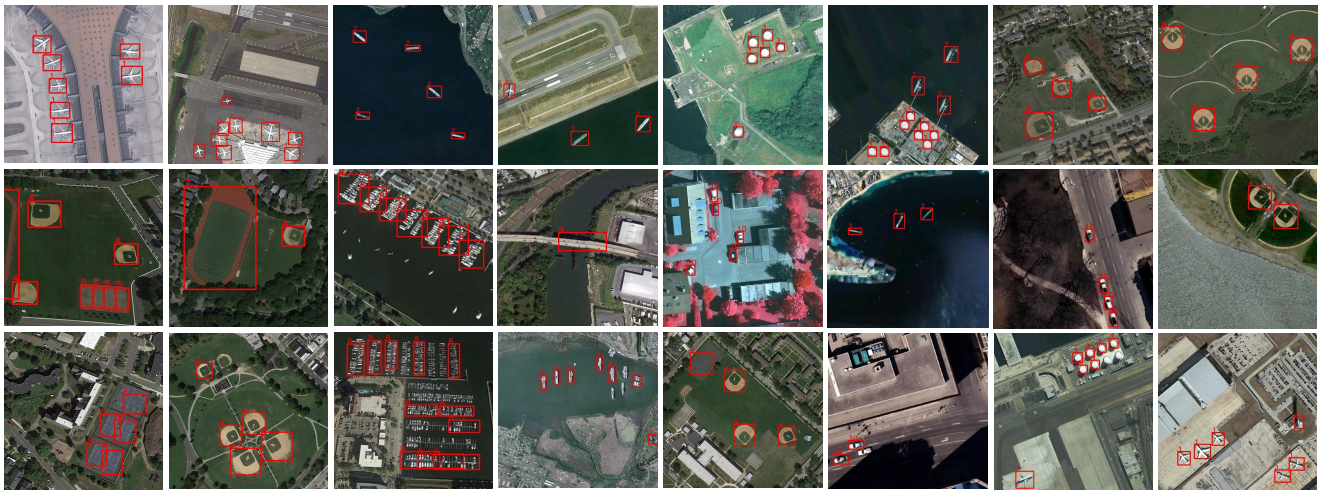

Fig. 7. Example results on the NWPU VHR-10 test split (73.1% mAP). The first two rows correspond to success cases. The third rows indicate a few missed objects and FPs. (categories: 1-airplane, 2-ship, 3-storage tank, 4-baseball diamond, 5-tennis court, 6-basketball court, 7-ground track field, 8-harbor, 9-bridge, and 10-vehicle).

The momentum is set as 0.9, which means the weight of the previous update gradient. In order to prevent overfitting and constrain the training model size, the Weight decay is set as 1e-4. Additionally, the initial learning rate is 0.01 and then reduced by a factor of 10 at iteration 60k. A total of 70k epochs are executed for our experiments on GeForce GTX 2080ti GPU.

C. Evaluation Metrics

The precision-recall curve (PRC) and average precision (AP), which are standard and authoritative evaluation indicators, are adopted to quantitatively evaluate different methods in remote sensing object detection.

1) Precision-Recall Curve: It is necessary to introduce the confusion matrix to evaluate different methods. The confusion matrix consists of four parts, which are true-positive (TP), true negative (TN), false-positive (FP), and false-negative (FN). Then, Precision and Recall are expressed as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{12}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{13}$$

Additionally, after checking the category, comparing location is inevitable in order to distinguish whether the predicted box is TP sample or FP sample. Intersection-over-union (IoU) between the ground truth (GT) and the predicted result (PR) is a crucial measurement for comparing location, which can be described as

$$\text{IoU} = \frac{\text{GT} \cap \text{PR}}{\text{GT} \cup \text{PR}}. \tag{14}$$

To be specific, the predicted boxes with IoU bigger than 0.5 are sentenced to be positive and the rest are negative.

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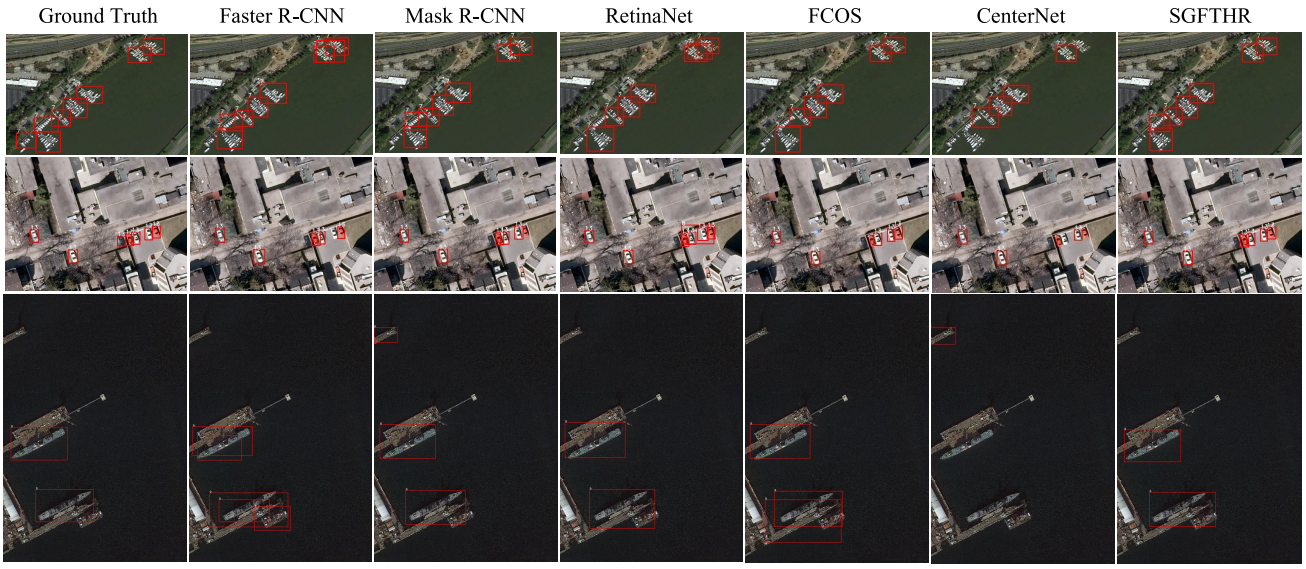

Fig. 8. Performances of our proposed SGFTHR network and the state-of-the-art methods. The first column and the last column present the GT and the detection results of the SGFTHR network, respectively, while others show the results of the state-of-the-art methods. (categories: 7-harbor, 8-parking lot, 9-ship, and 13-vehicle).

D. Experimental Results and Analysis

1) Average Precision: Generally, AP is the integration of the PRC from 0 to 1 on recall, which is a more comprehensive and robust indicator of the detection method performance. The larger the AP value, the higher the correct rate of object detection. On the contrary, the smaller the AP value, the higher the error rate of object detection. Moreover, mean AP (mAP) is the average of AP for all categories.

In this section, the performance of the two proposed modules in the SGFTHR network is evaluated quantitatively and qualitatively on the HRRSD dataset at first, including the SGFT module and the HR module. Then, the comparison between our proposed method and state-of-the-art methods on the HRRSD dataset and the NWPU VHR-10 dataset is shown. The detection results on different datasets are shown in Tables I–III. In the table, the bold data indicates the maximum, and the underlined data indicate the submaximal value.

2) Ablation Experiments: An ablation experiment with the same parameter settings on the HRRSD dataset is performed to analyze the SGFTHR network. The mAP is adopted as a comprehensive assessment. Additionally, the different role of each module is analyzed in our experiments.

The quantitative result of the ablation experiment is displayed in Table I. As we can see, the overall accuracy of detection assisted by the SGFT module acquires 1.9% performance gains in the aspect of mAP and small objects obtain larger improvements. Moreover, the improvements in ships and basketball courts are 2.54% and 6.51%. The vehicle, harbor, bridges, tennis court, parking lots, and T junction also make great progress, the specific improvements are 0.64%, 1.19%, 2.71%, 1.55%, 4.49%, and 3.17%, specifically. With the purpose of better comprehending representative structural information contained in the structure tensor more intuitively and clearly, the extraction effect of the structure tensor is visualized in Fig. 3. The visualization results indicate that the high-frequency components are sufficiently represented by the large eigenvalues in the structure tensor. The performance gains in the aspect of mAP assisted by the SGFT module imply that guiding the discriminative structural features contained in the structure tensor into high-level contextual feature maps contributes to a better object detector.

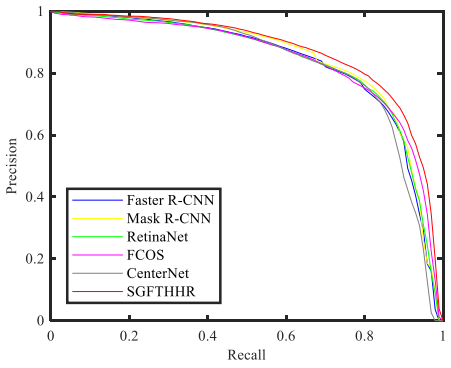

Fig. 9. PRCs of the proposed SGFTHR and other state-of-the-art methods for airplane, baseball diamond, basketball, bridge, crossroad, ground track field, harbor, parking lot, ship, storage tank, T junction, tennis court, and vehicle classes, respectively.

Besides, merely through embedding the HR module into the backbone, the model achieves an average 2.56% improvement in terms of mAP in the HRRSD dataset. It is worth noting that the objects with multiscale obtain larger gain. The improvements of the basketball court, T junction, parking lot, ship, and bridge are 6.19%, 5.26%, 6.14%, 3.30%, and 4.39%, respectively. As a result, we can conclude that the multiscale representation ability of the backbone is advanced at a granular level through constructing hybrid hierarchical residual-like connections with different kernel sizes in a single residual block.

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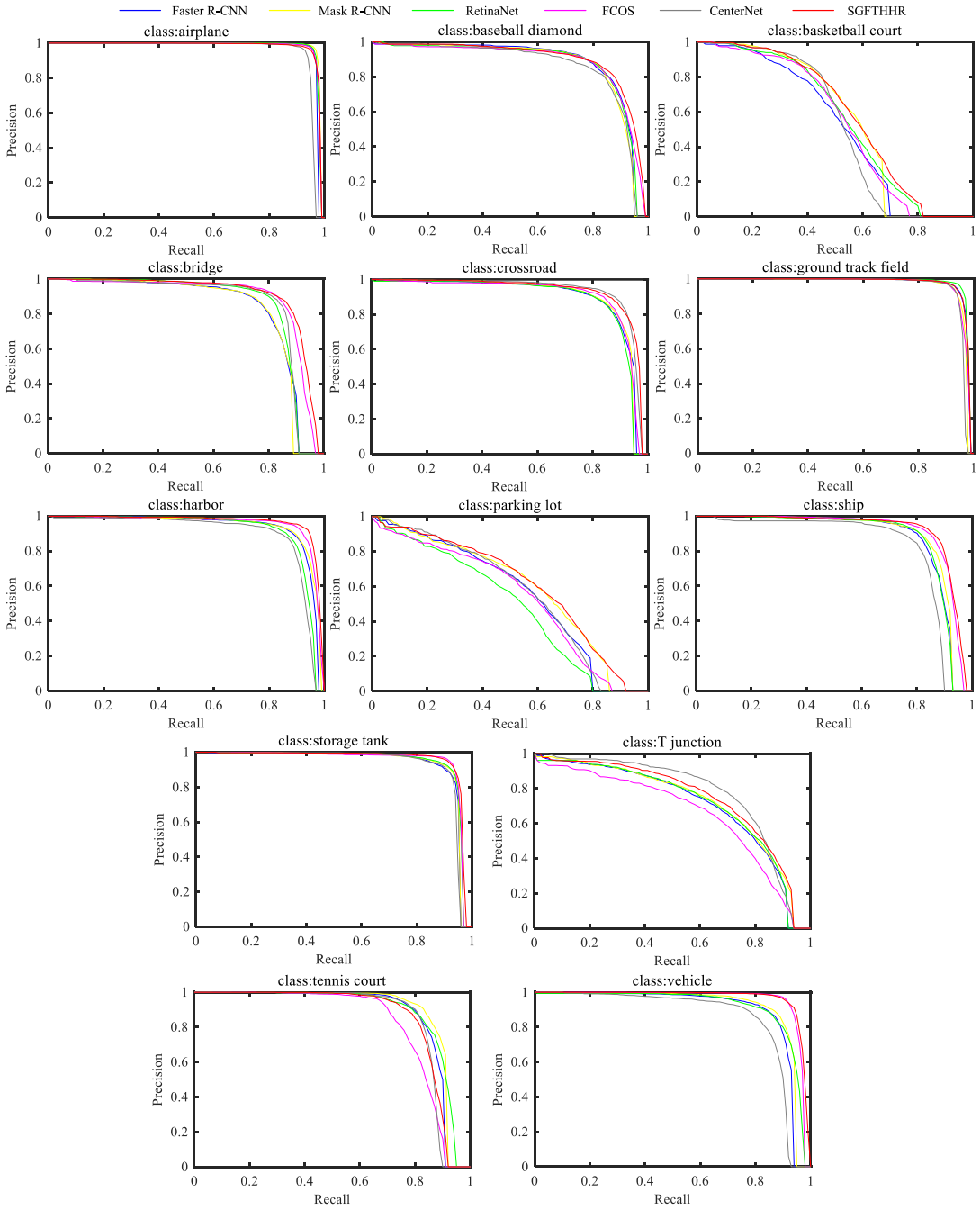

Fig. 10. PRCs of the proposed SGFTHR network and other state-of-the-art approaches on the HRRSD dataset.

The PRC curves of all objects and different categories in ablation experiments on the HRRSD dataset are displayed in Figs. 4 and 5, respectively. In the legend, "None" represents the baseline. "SGFT," "HR," and "SGFTHR" means the baseline with the SGFT module, the HR module, and both of them, respectively. Note that each of the modules plays a positive character in enhancing the detection accuracy in mAP, especially for small objects in RSI. The union of the SGFT module and the HR module greatly improves the overall accuracy of detection from 82.26% mAP to 86.38% mAP, which can mostly guide the network to focus on the discriminative structural feature and spatial details. Due to the two novel modules, the SGFTHR network obtains the discriminative structural feature representation ability and the multiscale spatial information enhancement ability to effectively recognize objects and boost the performance of small objects in spite of the diversity and complexity of background disturbing.

3) Comparison With the State-of-the-Art: We compare the detection performance of the SGFTHR network with state-of-the-art methods, including three one-stage methods and two two-stage methods. One-stage methods are the RetinaNet, the FCOS, and the CenterNet. Two-stage methods are Faster R-CNN and Mask R-CNN. Some examples of detection results in each class obtained by the SGFTHR network on the

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HRRSD dataset and the NWPU VHR-10 dataset are provided in Figs. 6 and 7, respectively. As we can see, the SGFTHR network has detected and located most of the objects successfully despite the large variations in the directions and sizes on the two datasets. The failure results are present in Figs. 6 and 7 (bottom row). The SGFTHR network demonstrates a poor performance on the classes of the basketball court and tennis court with false detection. Due to lacking instance-level supervision, the SGFTHR network sometimes misjudges the categories of the basketball court and tennis court that usually appear together in the same image.

To further demonstrate the effectiveness of our proposed network, the quantitative results of our method and the state-of-the-art detectors on the HRRSD dataset and the NWPU VHR-10 dataset are displayed in Tables II and III, respectively. The AP scores for 13 class objects intuitively indicate the detection performance of each algorithm. The SGFTHR network boosts the mAP score from 0.8417 to 0.8638 on the HRRSD dataset and from 0.7097 to 0.7310 on the NWPU VHR-10 dataset, which not only indicates the effectiveness of fully utilizing discriminative structural information but also implies the improvement of the multiscale representation ability in a hybrid hierarchical residual-like manner at a more granular level. As can be observed in Tables II and III, our method achieves the high accuracy of 86.38% mAP on the HRRSD dataset and 73.10% mAP on the NWPU VHR-10 dataset, which outperforms not only one-stage detectors, e.g., RetinaNet, CenterNet, and FCOS by 3.09%, 3.85%, and 4.12% on the HRRSD dataset and 4.79%, 5.32%, and 2.35% on the NWPU VHR-10 dataset but also two-stage detectors, e.g., Faster R-CNN and Mask R-CNN by 3.28% and 2.21% on the HRRSD dataset and 3.46% and 2.1% on the NWPU VHR-10 dataset. The effectiveness and robustness ability of the SGFTHR network is conclusively demonstrated by the thorough and convincing experiments. Based on the quantitative results, it is worth mentioning that our method displays a superior performance especially for small and dense objects, such as vehicles, parking lots, ships, harbors, and storage tanks, which illustrates that the characteristic structural features are well preserved and guided into the deep layers of the network. To prove the validity of our proposed SGFTHR network, a visualization of the detection results of our proposed SGFTHR network and the state-of-the-art detectors on the HRRSD dataset is present in Fig. 8. The first column and the last column present the GT and the detection results of the SGFTHR network, respectively, while others show the results of the state-of-the-art methods. This comparison between detection results yields two conclusions.

a) Guiding the structural features into the high-level features can reduce FPs: For instance, both the "parking lot" prediction for the "vehicle" scene and the "ship" prediction on the wharf are eliminated in the detection results of the proposed SGFTHR network.

b) Better multiscale feature representation can improve recall: For the harbor, vehicle, and ship classes, our proposed SGFTHR network detects more objects than does the other state-of-the-art methods.

Figs. 9 and 10 present the quantitative comparison results of six different methods performed on the HRRSD dataset measured by PRCs. However, although our proposed SGFTHR network outperforms these other methods (as illustrated in Fig. 9), the performance of the classes of T junction and tennis court is still poor. The main reason is that after the pooling operation is performed to extract features, some critical properties of these classes, for instance, the arc and straight lines in T junction and tennis courts, are reduced and even disappear. The decrease of these critical properties has prominently reduced the detection accuracy for those categories characterized by line properties.

V. Conclusion

In this article, we present the SGFTHR network to guide discriminative structural features into high-level contextual feature maps and exploit the multiscale potential at a more granular level. The discriminative structural features obtained by the structure tensor extractor are guided into high-level contextual feature maps to prevent the low-level structural information from being lost when the network goes deeper. The multiscale representation ability of the SGFTHR network is advanced at a more granular level through constructing HR-like connections with different kernel sizes in a single residual block. The adequate experiments are performed on the HRRSD dataset and the NWPU VHR-10 dataset, which not only illustrates the effectiveness and superiority of the SGFTHR network but also provides new insights for other researchers to address the task of object detection in RSI.

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References

[1] L. Zhou, J. Liu, and L. Chen, "Vehicle detection based on remote sensing image of YOLOv3," in *Proc. IEEE 4th Inf. Technol., Netw., Electron. Autom. Control Conf. (ITNEC)*, vol. 1, Jun. 2020, pp. 468–472.

[2] D. Pang, T. Shan, W. Li, P. Ma, and R. Tao, "Infrared dim and small target detection based on greedy bilateral factorization in image sequences," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, no. 99, pp. 3394–3408, Jun. 2020.

[3] X. Zhao *et al.*, "Joint classification of hyperspectral and LiDAR data using hierarchical random walk and deep CNN architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7355–7370, Oct. 2020.

[4] Y. Zhang, W. Li, H.-C. Li, R. Tao, and Q. Du, "Discriminative marginalized least-squares regression for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3148–3161, May 2020.

[5] H. Su, Z. Wu, Q. Du, and P. Du, "Hyperspectral anomaly detection using collaborative representation with outlier removal," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 12, pp. 5029–5038, Dec. 2018.

[6] C. Ge, Q. Du, W. Sun, K. Wang, J. Li, and Y. Li, "Deep residual network-based fusion framework for hyperspectral and LiDAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2458–2472, 2021.

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[7] T. Jiang, W. Xie, Y. Li, and Q. Du, "Discriminative semi-supervised generative adversarial network for hyperspectral anomaly detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2020, pp. 2420–2423.

[8] W. Xie, X. Zhang, Y. Li, K. Wang, and Q. Du, "Background learning based on target suppression constraint for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5887–5897, 2020.

[9] H. Guo, X. Yang, N. Wang, B. Song, and X. Gao, "A rotational libra R-CNN method for ship detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5772–5781, Aug. 2020.

[10] K. Jiang, W. Xie, Y. Li, J. Lei, G. He, and Q. Du, "Semisupervised spectral learning with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 5224–5236, Jul. 2020.

[11] Z. Zhang, R. Jiang, S. Mei, S. Zhang, and Y. Zhang, "Rotation-invariant feature learning for object detection in VHR optical remote sensing images by double-net," *IEEE Access*, vol. 8, pp. 20818–20827, 2020.

[12] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.

[13] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[14] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 91–99.

[15] K. He, G. Gkioxari, P. Dollar, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2961–2969.

[16] J. Dai, Y. Li, K. He, and J. Sun, "R-FCN: Object detection via region-based fully convolutional networks," in *Proc. NIPS*, 2016, pp. 379–387. [Online]. Available: http://papers.nips.cc/paper/6465-r-fcn-object-detection-via-region-based-fully-convolutional-networks

[17] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 779–788.

[18] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 7263–7271.

[19] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*. [Online]. Available: https://arxiv.org/abs/1804.02767

[20] A. Bochkovskiy, C.-Y. Wang, and H.-Y. Mark Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*. [Online]. Available: http://arxiv.org/abs/2004.10934

[21] W. Liu *et al.*, "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 21–37.

[22] C.-Y. Fu, W. Liu, A. Ranga, A. Tyagi, and A. C. Berg, "DSSD: Deconvolutional single shot detector," 2017, *arXiv:1701.06659*. [Online]. Available: http://arxiv.org/abs/1701.06659

[23] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 734–750.

[24] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6568–6577.

[25] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9627–9636.

[26] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[27] D. M. McKeown and J. L. Denlinger, "Cooperative methods for road tracking in aerial imagery," in *Proc. Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Apr. 1988, pp. 662–663.

[28] E. P. Baltsavias, "Object extraction and revision by image analysis using existing geodata and knowledge: Current status and steps towards operational systems," *ISPRS J. Photogramm. Remote Sens.*, vol. 58, nos. 3–4, pp. 129–151, Jan. 2004.

[29] T. Blaschke, "Object based image analysis for remote sensing," *ISPRS J. Photogramm. Remote Sens.*, vol. 65, no. 1, pp. 2–16, 2010.

[30] K. Hu *et al.*, "Vision-based freezing of gait detection with anatomic directed graph representation," *IEEE J. Biomed. Health Informat.*, vol. 24, no. 4, pp. 1215–1225, Apr. 2020.

[31] X. Shang *et al.*, "Target-constrained interference-minimized band selection for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6044–6064, Jul. 2021.

[32] B. Xue *et al.*, "A subpixel target detection approach to hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5093–5114, Sep. 2017.

[33] J. Lei, S. Fang, W. Xie, Y. Li, and C.-I. Chang, "Discriminative reconstruction for hyperspectral anomaly detection with spectral learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7406–7417, Oct. 2020.

[34] Y. Wang *et al.*, "A posteriori hyperspectral anomaly detection for unlabeled classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3091–3106, Jun. 2018.

[35] M. Song, F. Li, C. Yu, and C.-I. Chang, "Sequential band fusion for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, early access, Apr. 8, 2021, doi: 10.1109/TGRS.2021.3069734.

[36] W. Xie, Y. Li, J. Lei, J. Yang, C.-I. Chang, and Z. Li, "Hyperspectral band selection for spectral–spatial anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3426–3436, May 2020.

[37] G. Zhang, D. Zhu, W. Shi, X. Ye, J. Li, and X. Zhang, "Multi-dimensional residual dense attention network for stereo matching," *IEEE Access*, vol. 7, pp. 51681–51690, 2019.

[38] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[39] K. Li, G. Cheng, S. Bu, and X. You, "Rotation-insensitive and context-augmented object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2337–2348, Apr. 2018.

[40] Y. Zhong, X. Han, and L. Zhang, "Multi-class geospatial object detection based on a position-sensitive balancing framework for high spatial resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 138, pp. 281–294, Apr. 2018.

[41] W. Guo, W. Yang, H. Zhang, and G. Hua, "Geospatial object detection in high resolution satellite images based on multi-scale convolutional neural network," *Remote Sens.*, vol. 10, no. 1, p. 131, Jan. 2018.

[42] L. Liu, Z. Pan, and B. Lei, "Learning a rotation invariant detector with rotatable bounding box," 2017, *arXiv:1711.09405*. [Online]. Available: http://arxiv.org/abs/1711.09405

[43] W. Liu, L. Ma, and H. Chen, "Arbitrary-oriented ship detection framework in optical remote-sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 6, pp. 937–941, Jun. 2018.

[44] A. J. Collegio, J. C. Nah, P. S. Scotti, and S. Shomstein, "Attention scales according to inferred real-world object size," *Nature Hum. Behav.*, vol. 3, no. 1, pp. 40–47, 2019.

[45] C. G. Harris and M. Stephens, "A combined corner and edge detector," in *Proc. Alvey Vis. Conf.*, vol. 15, no. 50, 1988, pp. 10–5244.

[46] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[47] J. Gleason, A. V. Nefian, X. Bouyssounousse, T. Fong, and G. Bebis, "Vehicle detection from aerial imagery," in *Proc. IEEE Int. Conf. Robot. Automat.*, May 2011, pp. 2065–2070.

[48] K. Liu and G. Mattyus, "Fast multiclass vehicle detection on aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 9, pp. 1938–1942, Sep. 2015.

[49] T. N. Mundhenk, G. Konjevod, W. A. Sakla, and K. Boakye, "A large contextual dataset for classification, detection and counting of cars with deep learning," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 785–800.

[50] H. Zhu, X. Chen, W. Dai, K. Fu, Q. Ye, and J. Jiao, "Orientation robust object detection in aerial images using deep convolutional neural network," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2015, pp. 3735–3739.

[51] Z. Liu, H. Wang, H. Weng, and L. Yang, "Ship rotated bounding box space for ship extraction from high-resolution optical satellite images with complex backgrounds," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 8, pp. 1074–1078, Aug. 2016.

[52] Y. Zhang, Y. Yuan, Y. Feng, and X. Lu, "Hierarchical and robust convolutional neural network for very high-resolution remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5535–5548, Aug. 2019.

[53] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[54] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9759–9768.

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[55] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 770–778.

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